📄 NEXUS 4.0 — Technical Formulas

Golden Protocol Nexus

Technical Formulas and Protocol Invariants

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Date: 13.04.2026

Abstract

This document defines the core mathematical invariants underlying the Golden Protocol Nexus architecture.

These formulas do not introduce new mechanisms, but provide a formal representation of constraints and relationships already inherent in the protocol design, as described in the Founder Paper.

Their purpose is to explicitly express the structural properties governing supply, order reconstruction, and issuance conditions.

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1. Supply Constraint

$$S(t) \leq VRS(t)$$

Where:

- S(t) represents the circulating supply at time t
- VRS(t) represents the Verified Reserve State at time t

Definition:

At any time, the circulating supply cannot exceed the verified reserves recognized by the protocol.

This invariant ensures that asset-backed integrity is preserved and prevents inflation beyond validated custody.

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2. Order Reconstruction

$$R = \sum_{i=1}^{n} s_i$$

Where:

- R represents the total order
- s_i represents the i-th shard
- n represents the total number of shards

Definition:

Each order is fragmented into independent shards and subsequently reconstructed as the sum of all validated shard executions.

This invariant formalizes the compositional nature of order execution within the protocol.

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3. Issuance Condition

$$I(s_i) = 1 \Leftrightarrow \mathrm{Valid}(s_i)$$

Where:

- $I(s_i)$ is the issuance function for shard s_i
- $\mathrm{Valid}(s_i)$ indicates that the shard satisfies all protocol validation conditions

Definition:

Issuance is triggered if and only if the corresponding shard satisfies all required validation steps, including custody verification, audit confirmation, and protocol state consistency.

This invariant ensures that issuance is not discretionary, but strictly derived from valid system state.

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4. Conceptual Note

The formulas above represent a formal abstraction of protocol behavior.

They do not extend or modify the underlying architecture described in the Founder Paper, but clarify its logical structure by expressing:

- supply constraints
- compositional reconstruction
- state-dependent issuance

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5. Authorship and Timestamp Intent

This document is intended to establish authorship and formal definition of the above invariants.

A cryptographic timestamp (e.g., anchored on Bitcoin) may be associated with this document to provide immutable proof of existence at a given time.

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